

Chapter-3

RELATIVE VELOCITY

- It is the vector difference between actual velocity of two bodies

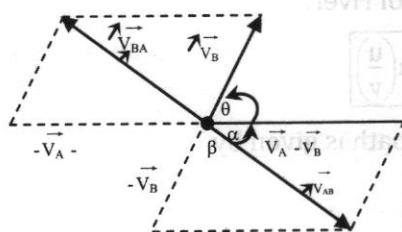
$$\vec{V}_{AB} = \text{Velocity of body A with Respect to body B}$$

OR

= Velocity of body 'A' relative to 'B'

$$\vec{V}_{BA} = \text{velocity of body 'B' with respect to body 'A'}$$

$$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B \text{ and } \vec{V}_{BA} = \vec{V}_B - \vec{V}_A$$



$$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$$

$$|\vec{V}_{AB}| = \sqrt{V_A^2 + V_B^2 - 2V_A V_B \cos \theta}$$

If $\vec{V}_{AB}$ makes an angle ' α ' with $\vec{V}_A$ then

$$\tan \alpha = \left(\frac{V_B \sin \theta}{V_A - V_B \cos \theta} \right)$$

$$\tan \beta = \left(\frac{V_A \sin \theta}{V_B - V_A \cos \theta} \right) \text{ from } (-\vec{V}_B)$$

- When two bodies are moving in same direction, then

$$|\vec{V}_{AB}| = |V_A - V_B| \quad (\theta = 0^\circ)$$

- When two bodies moves perpendicularly then

$$|\vec{V}_{AB}| = \sqrt{V_A^2 + V_B^2} \quad (\theta = 90^\circ)$$

- When two bodies are moving in opposite direction then

$$|\vec{V}_{AB}| = (V_A + V_B) \quad (\theta = 180^\circ)$$

- When two bodies 'A' and B are of a separation of 'd'.

- When body 'A' approaches to body 'B' moving in same direction then.

$$\text{a. Rate of approaching} = (V_A - V_B)$$

$$\text{b. Relative separation after time 't'}$$

$$x_t = d - (V_A - V_B) t$$

$$\text{c. time taken to overtake is given by}$$

$$t = \frac{d}{(V_A - V_B)}$$

- When both bodies approaches each other (ie. moving in apposite direction)

$$\text{a. Rate of approaching} = (V_A + V_B)$$

$$\text{b. Relative separation after time 't' } x_t = d - (V_A + V_B) t$$

$$\text{c. Time taken to cross ie, time of crossing is given by.}$$

$$t = \frac{d}{(V_A + V_B)}$$

Motion Across- River

$\vec{V}_{mw} = \vec{V} \Rightarrow$ It is the velocity of man, with which he can swim in Still water.

It is the capacity of swimmer.

$\vec{V}_w = \vec{u} \Rightarrow$ It is the velocity of water in the river

$\alpha \Rightarrow$ It is the direction of head of man with the direction of flow.

Or

It is the angle with downstream.

D= width of river.

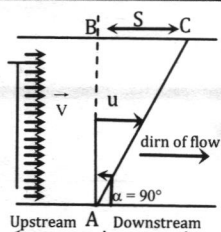
S = Down stream drift $\Rightarrow$ It is that distance along the direction of flow by which swimmer shifts from original point.

$\Rightarrow$ Time taken by swimmer to travel distance 'l'

$$\text{a. In upstream of the river of given by, } t = \frac{l}{v-u}$$

$$\text{b. In down stream of the river is given by } t = \frac{l}{v+u}$$

Shortest time



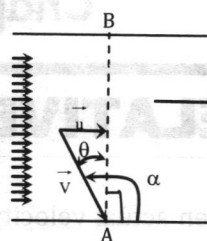
→ for shortest time, i.e, minimum time swimmer should start swimming perpendicular to the direction of flow

1. $\alpha = 90^\circ$

2. $t_{\text{shortest}} = \frac{D}{v}$

3. $S = \frac{uD}{v}$

Shortest path



→ $\sin\theta = \frac{u}{v} \Rightarrow u = v\sin\theta \therefore \sin\theta < 1$

$\frac{u}{v} < 1$

⇒ $v > u$ ⇒ This is the required condition for swimmer to reach direct opposite point (i.e. along the shortest path on the another bank of river.

2. $\alpha = 90 + \theta = 90 + \sin^{-1}\left(\frac{u}{v}\right)$

3. Time along shortest path is given by

$t = \frac{d}{v\cos\theta} = \frac{D}{\sqrt{v^2 - u^2}}$

⇒ Time taken by swimmer to travel distance 'l' in upstream or in downstream and return back is given by.

$t = \frac{l}{v-u} + \frac{l}{v+u} = \frac{2lv}{v^2 - u^2}$

⇒ If t_1 and t_2 are the time taken by swimmer to travel same distance in upstream and down stream respectively then time taken by him to travel same distance in still water is given by:

$t = \frac{2t_1t_2}{t_1+t_2}$

Motion of Rain drops and motion of winds

$\vec{V}_r \Rightarrow$ It is the velocity of rain drops observed by man when he is stationary. Or it is the actual velocity of Raindrops

$\vec{V}_{rm} \Rightarrow$ It is velocity of raindrops observed by man when he is moving.

$\vec{V}_m \Rightarrow$ It is velocity of man relative to ground.

$\vec{V}_w \Rightarrow$ It is the actual velocity of wind or it is the velocity of wind observed by man when he is stationary.

$\vec{V}_{wm} \Rightarrow$ It is the velocity of wind observed by man when he is moving.

i. $\vec{V}_{rm} = \vec{V}_r - \vec{V}_m$ and $\vec{V}_{wm} = \vec{V}_w - \vec{V}_m$

$|\vec{V}_{rm}| = \sqrt{V_r^2 + V_m^2 - 2V_rV_m\cos\theta}$

And $|\vec{V}_{wm}| = \sqrt{V_w^2 + V_m^2 - 2V_wV_m\cos\theta}$

ii. $\vec{V}_r = \vec{V}_{rm} + \vec{V}_m$ and $\vec{V}_w = \vec{V}_{wm} + \vec{V}_m$

$|\vec{V}_r| = \sqrt{V_{rm}^2 + V_m^2 + 2V_rV_m\cos\theta}$

$|\vec{V}_w| = \sqrt{V_{wm}^2 + V_m^2 + 2V_wV_m\cos\theta}$

Note: Both case are analyzed by similar method

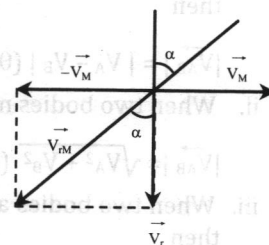
Case I: When man is stationary:

When a man is stationary rain strikes him vertically. Now he starts moving then,

$\vec{V}_{rm} = ?$

$\vec{V}_{rm} = \vec{V}_r - \vec{V}_m$

Angle between



$$\vec{V}_r \text{ and } \vec{V}_m$$

$$\theta = 90^\circ$$

$$|\vec{V}_{rm}| = \sqrt{V_r^2 + V_m^2}$$

$$\tan \alpha = \left(\frac{V_m}{V_r} \right) \text{ from vertical or from } \vec{V}_r$$

Case II: When a man is moving rain appears to strike him vertically then actual velocity of rain drops is given by.

a. Magnitude $|\vec{V}_r| = \sqrt{V_{rm}^2 + V_m^2} [\theta = 90^\circ]$

b. Direction: $\tan \alpha = \left(\frac{V_m}{V_{rm}} \right)$ From vertical

Case III: When a man is moving with velocity $\vec{V}_m$ rain appears to strike him vertically. When his velocity becomes $\vec{V}_m^1$ ($V_m^1 = nV_m$) then rain appears to strike him of angle ' θ ' with vertical then actual velocity of rain drop is given by

a. Direction:

$$\cot \alpha = (n-1) \cot \theta = \left(\frac{V_m^1}{V_m} - 1 \right) \cot \theta$$

b. Magnitude: $|\vec{V}_r| = \frac{V_m}{\sin \alpha}$

Motion of plane

$\vec{V}_{pw} \Rightarrow$ It is velocity of plane relative to wind.

$\vec{V}_w =$ It is the velocity of wind

$\vec{V}_p =$ Actual velocity of plane or it is the velocity of plane relative to ground.

$$\vec{V}_{pw} = \vec{V}_p - \vec{V}_w \quad \text{and} \quad \vec{V}_p = \vec{V}_{pw} + \vec{V}_w$$

Multiple Choice Questions

1. A train of length 150m is moving towards north with velocity 10 m/sec. A parrot is flying with velocity 5m/sec towards south direction parallel to railway track. The time taken by the parrot to cross the train is:

- a. 15 sec b. 10sec
c. 30sec d. 12sec

2. Two trains each of length 50m are moving in opposite direction towards each other on parallel lines with speeds 10m/sec. then the time of crossing will be.

- a. 2sec b. 4sec
c. 5sec d. 10sec

3. To a person going east in a car with velocity of 25km/hr a train appears to move towards north with a velocity of $25\sqrt{3}$ km/hr. The actual velocity of the train will be

- a. 25km/hr b. $25\sqrt{2}$ km/hr
c. 50 km/hr d. $50\sqrt{3}$ km/hr

4. Two cars are moving in same direction with the same speed of 30 km/hr. They are separated by a distance 5km. A third car moving in opposite direction meets these two cars at an interval of 4 minutes. The speed of the car will be ?

- a. 105 km/hr b. 75 km/hr
c. 60 km/hr d. 45 km/hr

5. A bird is flying towards south with a velocity of 40km/hr and a train is moving with velocity 40km/hr towards east. What is the velocity of the bird w.r.t an observer in train

- a. $40\sqrt{2}$ km/hr N-E
b. $40\sqrt{2}$ km/hr S-E
c. $40\sqrt{2}$ Km/hr S-W
d. $40\sqrt{2}$ km/hr N-W

6. Two cars each of length 20m are moving in such a way that their time of crossing is 2 sec and their time of overtaking is 4sec then speeds of the can will be.

- a. 10 m/sec and 5 m/sec
b. 20mlsec and 10ml/sec
c. 15 m/sec and 5 m/sec
d. 15m/sec and 10 m/sec

7. A car X is moving with velocity 80 km/hr towards North-east direction and another car y moving with velocity of 60km/hr towards south-east direction. Then the angle made by velocity of car X relative to Car y from North will be.

- a. $\tan^{-1}\left(\frac{1}{7}\right)$ b. $\tan^{-1}\left(\frac{2}{7}\right)$
c. $\tan^{-1}\left(\frac{3}{4}\right)$ d. $\tan^{-1}\left(\frac{4}{3}\right)$

8. Wind is blowing west to east. Two trains moving with same speed along two -parallel tracks in opposite direction have the stream of track of one

is doubled than other. The speed of each train will be:

- equal to that of wind
- double that of wind
- half that of wind
- three times that of wind

9. A police patrol Jeep is pursuing a car in straight level road. The jeep and the car are moving with speeds of 60km/hr and 72km respectively. The patrol jeep fires a bullet when the car is 200m from it. If the speed of bullet is 500m/sec, then the relative separation between the two vehicles when the bullet strikes the car will be.

- 198.66m
- 201.34m
- 198.6m
- 201.4m

10. Two cars A and B are moving in the same direction with the velocities V_1 and V_2 ($V_1 > V_2$). When the separation between the cars is 'd' the driver of A applies brakes and produces constant retardation 'a' then there will be no collision if.

- $d < \frac{(V_1 - V_2)^2}{2a}$
- $d < \frac{(V_1^2 - V_2^2)}{2a}$
- $d > \frac{(V_1 - V_2)^2}{2a}$
- $d > \frac{V_1 - V_2}{a}$

11. A car travelling at 60 km/hr overtakes another car travelling at 42km/hr. Assuming each car to be 5m long, find the time taken during the overtake.

- 1sec
- 2 sec
- 3sec
- 4sec

12. In above question the total road distance used for the overtakes?

- 10m
- 28.33 m
- 38.33m
- 18.33

13. A police jeep is chasing a culprit going on a motorbike. The motorbike crosses a turning of a speed of 72 km/hr. The jeep follows it of a speed of 90 km/hr Crossing the turning ten second alter than the bike. Assuming that they travel of constant speeds, then how far from the turning with the jeep catch up with the bike?

- 1000m
- 1250m
- 250m
- 500m

14. Stationary man observes that the rain is falling vertically downward. When he starts running with a velocity of 12 km/hr he observes that the rain

falling at an angle 60° with the vertical. The actual velocity of rain is

- $12\sqrt{3}$ km/hr
- 12 km/hr
- $4\sqrt{3}$ km/hr
- $10\sqrt{3}$ km/hr

15. A car is moving on a road and rain is falling vertically, then select the correct answer.

- The rain will strike the hind screen only
- The rain will strike the front screen only
- The rain will strike both the screen
- The rain will not strike any of the screen

16. A boatman wants to take his boat to appoint just opposite on the other bank of river. Then

- he should row up-stream
- he should row down -stream
- he should inclined to that bank of up-stream
- he should inclined to the bank down- stream.

17. A man is walking due east at the rate of 4km/hr and rain appears to him to be falling vertically. When he doubles his speed it appears to be falling at an angles of 45° with the vertical, then the actual speed and direction of the rain will be.

- $4\sqrt{2}$ km/hr, 45°
- 4km/hr, 45°
- $4\sqrt{2}$ km/hr, 90°
- $2\sqrt{2}$ km/hr, 0°

18. A boat which has a speed of 5km/hr, in still water crosses a river of width 1km along the shortest possible path in 15 minutes. The velocity of the river water will be

- 1 km/hr
- 4km/hr
- 3km/hr
- $\sqrt{41}$ km/hr

19. A river is flowing from west to east at a speed of 5m/min. A man on the south bank of the river capable of swimming at 10m/min in Stillwater wants to cross the river in shortest time, then in which direction he should starts swimming.

- 30° North of west
- 30° west of north
- due north
- 60° north of east

21. A boat takes 2hr to travel 8km and returning back in still water. If a river is flowing with velocity 4km/hr, then time taken by boat to travel 8km and returning back in upstream of the river will be?

- 2hr
- 2hr 40min
- 1hr 20min
- date is not sufficient

22. A person aiming to reach directly opposite point on the another bank of river capable of swimming

with velocity 0.5 m/sec. his head makes 120° with the direction of flow then the flow velocity of water in the river will be.

- a. 1m/sec b. 0.433 m/sec
c. 0.25m/sec d. 0.125m/sec

23. A man swims 3km in 3hr in still water, while a log floating downstream covers a distance of 1km during this time. How many km will the man swim upstream in 3hrs?

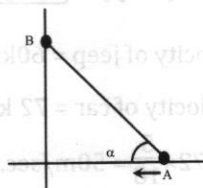
- a. 1m b. 2km
c. 3km d. 4m

24. A man can swim with a velocity of 5km/hr in still water, jumps into river of width 1km flowing with velocity 3km/hr perpendicular to the direction of flow. If he takes total of 20minute to reach directly apposite point on the another bank of river, then his velocity on the another band will be?

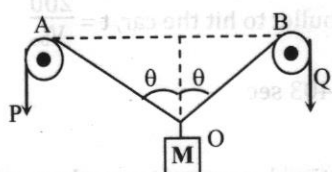
- a. 4.5 km/hr b. 9 km/hr
c. 5km/hr d. 1.8 km/hr

25. Two particles A and B are connected by a rigid rod AB the rod slides along perpendicular rails as shown. The velocity of A to the left is 20 m/sec. What is the velocity of 'B' when angles $\alpha=60^\circ$?

- a. 5.8 m/sec
b. 9.8 m/sec
c. 20 m/sec
d. 17.3 m/sec



26. In the arrangement shown in figure the ends 'P' and 'Q' of an unstretchable string move downwards with uniform speed 'u'. pulleys 'A' and B are fixed. The mass 'M' moves with a velocity of:



- a. $2 u \cos \theta$ b. $\frac{u}{\cos \theta}$
c. $\frac{2u}{\cos \theta}$ d. $u \cos \theta$

27. For a cyclist running 5m/sec due west, the rain appears to fall vertically. When he increases his speed to 17 m/sec due west, the rain appears to fall at an angle of 45° with the vertical. Find the actual speed of the rain.

- a. 12 m/sec b. 13 m/sec
c. 22m/sec d. 15m/sec

28. An airplane pilot wishes to fly due north. A wind 30 km/hr is blowing towards west. If the flying speed of the plane is 360 km/hr then what is the speed of plane over the ground.

- a. 359.6 km/hr b. 361.25 km/hr
c. 330 km/hr d. 390 km/hr

29. In flying an airplane, a pilot wants to maintain a velocity of 500 km/hr eastward with respect to ground velocity (magnitude and direction) must pilot maintain with respect to wind to do achieve the desired ground velocity?

- a. 502 km/hr, $\tan^{-1} \frac{2}{25}$ N of E
b. 502 km/hr, $\tan^{-1} \frac{2}{25}$ E of N
c. 540 km/hr, $\tan^{-1} \frac{2}{25}$ W of N
d. 540 km/hr, $\tan^{-1} \frac{2}{25}$ N of W

30. Three particles are placed at the corner of equilateral triangle having side 'a'. Each particle moves with constant speed 'v' in such a way that $1 \rightarrow 2$; $2 \rightarrow 3$; and $3 \rightarrow 1$ then after what time they will meet.

- a. $\frac{a}{v}$ b. $\frac{2a}{3v}$
c. $\frac{2a}{v}$ d. $\sqrt{\frac{2}{3}} \frac{a}{v}$

Answers:

1.b	2.b	3.c	4.d	5.c
6.c	7.a	8.d	9.b	10.c
11.b	12.c	13.a	14.c	15.b
16.c	17.a	18.c	19.c	20.b
21.b	22.c	23.d	24.a	25.a
26.b	27.b	28.a	29.a	30.b

Hints and Solution:

1. Time of crossing $t = \frac{\text{separation}}{\text{Rate}} = \frac{150}{10+5} = \frac{150}{15} = 10\text{sec}$

2. Rate = $(15 + 10)_1$ and 25m/sec
Separation (d) = $50 + 50 = 100$
 $t = \frac{\text{separation}}{\text{Rate}} = \frac{100}{15+10} = 4\text{sec}$

3. V_{TC} = velocity of train relative to C_2
 V_T = actual velocity of train
 V_c = actual velocity of car
 $\vec{V}_{TC} = \vec{V}_T - \vec{V}_C$

$$\Rightarrow \vec{V}_T = \vec{V}_{TC} + \vec{V}_C$$

$$\therefore V_T = \sqrt{V_{TC}^2 + V_C^2} \text{ (Angle between } \vec{V}_{TC} \text{ and } \vec{V}_C \text{ is } 90^\circ)$$

$$= \sqrt{(25\sqrt{3})^2 + 25^2} = 50 \text{ km/hr}$$

4. Let V = velocity of third car
Rate of approaching = $V + 30$

$$T = \frac{\text{separation}}{\text{Rate}}$$

$$\Rightarrow \left(\frac{4}{60}\right) = \frac{5}{V+30}$$

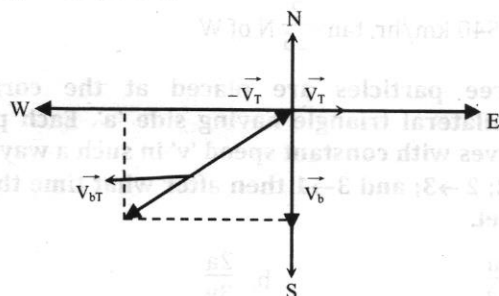
$$V+30=75$$

$$\therefore V=30 \text{ km/hr}$$

$$5. \vec{V}_{bT} = \vec{V}_b - \vec{V}_T = \vec{V}_b + (-\vec{V}_T)$$

$$|\vec{V}_{bT}| = \sqrt{V_b^2 + V_T^2} = 40\sqrt{2}$$

$$\therefore \vec{V}_{bT} = 40\sqrt{2} \text{ S-W}$$



6. Separation in each curve = sum of lengths of Cars
= $(20 + 20) = 40\text{m}$ Crossing occurs in apposite direction and overtaking occurs in same direction.

$$\text{Time of Crossing: } t_1 = \frac{40}{V_1 + V_2}$$

$$\Rightarrow V_1 + V_2 = 20 \dots (i)$$

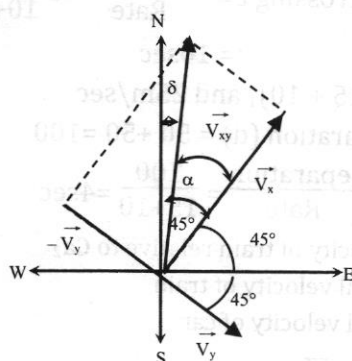
$$\text{Time of overtaking } t_2 = \frac{40}{V_1 - V_2}$$

$$\rightarrow V_1 - V_2 = 10 \rightarrow (ii)$$

Solving (i) and (ii) we get,

$$V_1 = 15 \text{ m/sec and } V_2 = 5 \text{ m/sec.}$$

7.



$$\vec{V}_{XY} = \vec{V}_X - \vec{V}_Y = \vec{V}_X + (-\vec{V}_Y)$$

If $\vec{V}_{XY}$ makes an angle ' α ' with $\vec{V}_X$ then

$$\tan \alpha = \frac{V_Y}{V_X} = \frac{60}{80} \text{ (Angle between } \vec{V}_Y \text{ and } \vec{V}_X \text{ is } 90^\circ)$$

$$= \tan \alpha = \frac{3}{4}$$

The angle made by $\vec{V}_{XY}$ with north (δ) is given by
 $\delta = 45^\circ - \alpha$

$$\tan \delta = \tan(45^\circ - \alpha) = \frac{\tan 45^\circ - \tan \alpha}{1 + \tan 45^\circ \tan \alpha} = \frac{1 - \frac{3}{4}}{1 + 1 \cdot \frac{3}{4}}$$

$$\Rightarrow \tan \delta = \frac{1}{7} \Rightarrow \delta = \tan^{-1}\left(\frac{1}{7}\right)$$

8. Let $\vec{V}_T = x$, and $\vec{V}_w = y$

For one train, $\vec{V}_{TW} = x - y \rightarrow (i)$

For another train $\vec{V}_{TW} = x + y \dots (ii)$

According to question

$$X = y = 2(x - y) = \boxed{x = 3y}$$

9. V_j = velocity of jeep = $60 \text{ km/hr} = 60 \times \frac{5}{18} = \frac{50}{3} \text{ m/sec}$

V_e = velocity of car = 72 km/hr

$$= 72 \times \frac{5}{18} = 50 \text{ m/sec.}$$

V_a = velocity of bullet relative to Jeep = 500 m/sec

V_b = Actual velocity of bullet

$$= \left(500 + \frac{50}{3}\right) = \frac{1550}{3} \text{ m/sec}$$

Time taken by bullet to hit the car, $t = \frac{200}{V_{bc}}$

$$T = \frac{200}{\left(\frac{1490}{3}\right)} = 0.403 \text{ sec}$$

Relative separation between Jeep and car when bullet hit the car

$$= 200 + (V_c - V_j) \times t = 200 + \left(20 - \frac{50}{3}\right) \times 0.403 = 201.34 \text{ m}$$

10. Initial Rate of approaching: $V = V_1 - V_2$

When velocity of car a becomes V_2 ie. equal to velocity of car B. find Rate of approaching $V = V_2 - V_2 = 0$

Using: $V^2 = U^2 + 2as$

$$0 = (v_1 - v_2)^2 + 2(-a) \times 5$$

$$S = \frac{(V_1 - V_2)^2}{2a} \text{ For no collision ;}$$

$$S < d \frac{(V_1 - V_2)^2}{2a} < d$$

$$\therefore d > \frac{(V_1 - V_2)^2}{2a}$$

11. Rate of approaching = $V_1 - V_2 = (60 - 42) = 18 \text{ km/hr} = 5 \text{ m/sec}$.

$$\text{Separation} = 5 \text{ m} + 5 \text{ m} = 10 \text{ m}$$

$$\text{Time of overtaking } (t) = \frac{60}{5} = 2 \text{ sec}$$

12. Total road distance used for the overtaking is giving by
d = length of car + distance moved by rear car during time of overtake.

$$= 5 \text{ m} + (60 \times \frac{5}{16}) \times 2 = 38.33 \text{ m}$$

Alternatively, d = length of both car + distance moved by rear car + length of car

$$= 5 + 5 + (42 \times \frac{5}{16}) \times 2 + 5 = 38.33 \text{ m}$$

13. Separation between jeep and bike is given by

$$S = \text{velocity of jeep} \times \text{time} = (90 \times \frac{5}{18}) \times 10 = 250 \text{ m}$$

$$\begin{aligned} \text{Rate of approaching} &= V_{\text{jeep}} - V_{\text{bike}} \\ &= (90 - 72) \times \frac{5}{18} = 5 \text{ m/sec} \end{aligned}$$

Time taken to catch the bike is given by.

$$T_p = \frac{\text{separation}}{\text{Rate}} = \frac{250}{5} = 50 \text{ sec}$$

Distance from turning is given by:

$$D = V_{\text{bike}} \times t = (72 \times \frac{5}{18}) \times 50 = 1000 \text{ m}$$

14. $\tan \alpha = (\frac{V_m}{V_r})$; from vertical

$$V_r = \frac{V_m}{\tan \alpha} = \frac{12}{\tan 60} = \frac{12}{\sqrt{3}} = 4\sqrt{3} \text{ km/hr}$$

17. $\text{Cot} \alpha = (n-1) \text{Cot} \theta$

$$= (2-1) \text{Cot} 45^\circ$$

$$\text{Cot} \alpha = \text{Cot} 45^\circ$$

$$\therefore \alpha = 45^\circ$$

Magnitude of actual velocity

$$V_r = \frac{V_m}{\sin \alpha} = \frac{4}{\sin 45^\circ} = 4\sqrt{2} \text{ km/hr.}$$

18. $V = 5 \text{ km/hr}$

$$\theta = 1 \text{ km}, t = 15 \text{ minute} = \frac{15}{60} \text{ hr} = \frac{1}{4} \text{ hr}$$

$$T = \frac{D}{\sqrt{V^2 - u^2}} \Rightarrow \frac{1}{4} = \frac{1}{\sqrt{5^2 - u^2}} \Rightarrow 25 - u^2 = 16$$

$$\therefore u = 3 \text{ km/hr}$$

19. For shortest time swimmer should start swimming perpendicular to flow, hence due North.

$$20. \sin \theta = \frac{u}{v} = \frac{5}{10} = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ \text{ from North to upstream}$$

$$\text{I.e. } \theta = 30^\circ \text{ West of north.}$$

$$21. V = \frac{16}{2} = 8 \text{ km/hr}$$

$$T = \frac{1}{v-u} + \frac{1}{v+u} = \left(\frac{8}{8-4} + \frac{8}{8+4} \right) \text{ hr} = 2 \text{ hr } 40 \text{ min}$$

$$22. \alpha = 120^\circ \text{ i.e. } \therefore \alpha = 90^\circ + \theta$$

$$\theta = \alpha - 90^\circ = 30^\circ$$

$$\sin \theta = \frac{u}{v}$$

$$\Rightarrow u = V \sin \theta = 0.5 \times \frac{1}{2} = 0.25 \text{ m/sec.}$$

$$23. V = \frac{3}{3} = 1 \text{ km/hr}$$

$$U = \frac{1}{3} \text{ km/hr}$$

$$\begin{aligned} \text{Upstream distance is given by; } l &= (v-u) \times t = (1 - \frac{1}{3}) \times 3 \\ &= 2 \text{ km} \end{aligned}$$

24. Since man jumps into river perpendicular to flow hence he takes minimum time to cross the river.

$$T = \frac{D}{v} = \frac{1}{5} \text{ hr}$$

$$\text{Downstream drift } S = \frac{uD}{V} = \frac{3 \times 1}{5} = 0.6 \text{ km}$$

$$\begin{aligned} \text{Time taken by man to reach directly opposite point} \\ \text{after crossing river } t' = t_{\text{total}} - t = d \left(\frac{20}{60} - \frac{1}{5} \right) = \frac{2}{15} \text{ hr} \end{aligned}$$

Speed on another bank

$$V' = \frac{S}{t'} = \frac{0.6}{(\frac{2}{15})} = 0.3 \times 15$$

$$V' = 45 \text{ km/hr}$$

25. Let $AB = l = \text{constant}$

$$V_A = \frac{dx}{dt} = 10 \text{ m/sec}$$

$$(\text{left}) \therefore V_A = -10$$

$$V_3 = \frac{dy}{dt}$$

$$X^2 + Y^2 = l^2$$

Differentiating

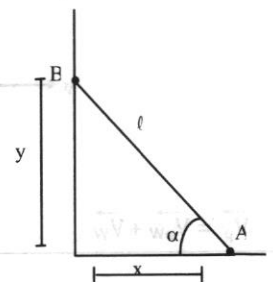
w.r.to. time.

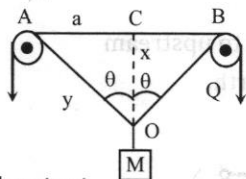
$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$\Rightarrow 2xV_A + 2yV_B = 0$$

$$V_B = -\frac{x}{y} V_A = -V_A \text{Cot} \alpha$$

$$\therefore V_B = -(-10) \times \frac{1}{\sqrt{3}} = \frac{10}{\sqrt{3}} = 5.8 \text{ m/sec (upwards)}$$



26. Let $OC = X$ and $OA = Y$

 $AC = a = \text{Constant}$
 $\frac{dx}{dt} = \text{velocity of mass 'M'}$
 $\frac{dy}{dt} = \text{velocity of ends 'p' \& 'Q' = u}$

$$\therefore a^2 + x^2 = y^2$$

Differentiating w.r.to time

$$0 + 2x \frac{dx}{dt} = 2y \frac{dy}{dt} \Rightarrow 2x V_m = 2y V_p$$

$$V_m = \frac{y}{x} V_p = \frac{u}{\cos \theta} \quad \left[\text{In given figure } \cos \theta = \frac{x}{y} \right]$$

 27. $V_m = 5$, $v'_m = 17$, $\theta = 45^\circ$

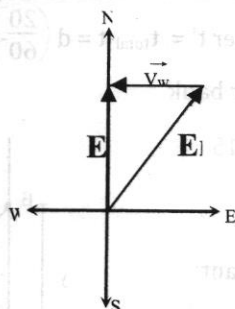
$$\cot \alpha = (n-1) \cot \theta = \left(\frac{V_m'}{V_m} - 1 \right) \cot \theta$$

$$\cot \alpha = \left(\frac{17}{5} \right) \cot 45^\circ = \frac{12}{5}$$

 $b = 12$; $P = 5$; $h = 13$

Magnitude of actual velocity;

$$V_r = \frac{V_m}{\sin \alpha} = \frac{5}{(5/13)} = 13 \text{ m/sec}$$

 28. $\vec{V}_w = 30/\text{hr}$ toward west


$$\vec{V}_p = \vec{V}_{pw} + \vec{V}_w$$

$$\vec{V}_{pw} = 360 \text{ km/hr}$$

From figure,

$$V_p^2 = V_{pw}^2 - V_w^2$$

$$V_p \sqrt{360^2 - 30^2} \quad [h^2 = [p^2 + b^2]]$$

$$\therefore V_p = 359.6 \text{ km/hr}$$

 29. $\vec{V}_p = 500 \text{ k/hr}$ (eastward)

$$\vec{V}_w = 40 \text{ km/hr}$$
 (south)

$$\vec{V}_{pw} = \vec{V}_p - \vec{V}_w$$

$$= \vec{V}_p + (-\vec{V}_w)$$

$$\tan \alpha = \left(\frac{V_w}{V_p} \right) = \frac{40}{500} = \left(\frac{2}{25} \right)$$

$$\alpha = \tan^{-1} \left(\frac{2}{25} \right) \text{ from north of}$$

East

$$|\vec{V}_{pw}| = \sqrt{(500)^2 + 40^2} = 501.6 = 502 \text{ km/hr}$$

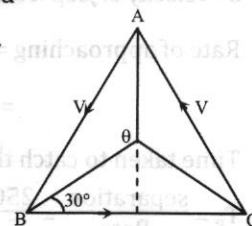
30. Three particles meet at Centroid

$$\Rightarrow OB = \frac{a/2}{\cos 30^\circ} = \frac{a}{2} \times \frac{2}{\sqrt{3}} = \frac{a}{\sqrt{3}}$$

Component of velocity along

$$OB = V \cos 30^\circ = \frac{\sqrt{3}V}{2}$$

$$\text{Time taken: } t = \frac{a/\sqrt{3}}{\left(\frac{\sqrt{3}V}{2} \right)} = \frac{2a}{3V}$$


 Alternative: Velocity of B is ' V ' along BC. The velocity of C is along CA. Its component along CB is $V \cos 60^\circ = V/2$. Thus the separation BC decreases at the rate

$$V + \frac{V}{2} = \frac{3V}{2}$$

Since, this rate is constant, the time taken in reducing the separation BC from a zero is

$$T = \frac{a}{\left(\frac{3V}{2} \right)} = \frac{2a}{3V}$$

